Power flags (ERC drivers for rails that enter through connectors)

GND

Main rails +/−28 V (T1)

J1 T1 SEC 20-0-20 VAC

BR1 GBJ2506

C1 10000u 50V

C2 10000u 50V

C3 10000u 50V

C4 10000u 50V

C5 1u 100V film

C6 1u 100V film

R1 4k7 0.5W

R2 4k7 0.5W

J2 TO POWER AMP +28/GND/−28

J3 AC SENSE to power amp

Star ground: GND net meets at the C1–C4 junction. Heavy tracks / wires to the power amp board.

−15 V regulator (from −28 V)

R21 10 1W

C21 1000u 50V

R22 2k2

D21 1N4745A 16V

R23 1k

C22 100u 25V

R24 1k

Q4 MPSA92

Q5 BD140

R25 2R7

Q6 MPSA92

C23 100u 25V

C24 100n

R26 4k7

J4 TO PREAMP +15/GND/−15

Mirror of the +15 V regulator.

+40 V regulator (discrete)

R31 4k7 0.5W

D35 1N4755A 43V

R32 10k

C35 100u 63V

R33 1k

Q11 MPSA42

Q12 MJE340

R34 12

Q13 MPSA42

C36 100u 63V

C37 100n 100V

R35 10k 0.5W

43 V zener ref, RC soft start. MPSA42 + MJE340 Darlington, Q13 limits at ~55 mA. Output ~ +41.6 V.

+15 V regulator (from +28 V)

R11 10 1W

C11 1000u 50V

R12 2k2

D11 1N4745A 16V

R13 1k

C12 100u 25V

R14 1k

Q1 MPSA42

Q2 BD139

R15 2R7

Q3 MPSA42

C13 100u 25V

C14 100n

R16 4k7

Ref: 16 V zener, RC-filtered. Darlington Q1/Q2. Q3 limits at ~240 mA. Output ~ +14.8 V.

HV raw supply (T2)

J5 T2 SEC 40-0-40 VAC

D31 1N4007

D32 1N4007

D33 1N4007

D34 1N4007

C31 1000u 100V

C32 1000u 100V

C33 100n 250V

C34 100n 250V

1N4007 bridge, 1000 uF 100 V per rail: ~ +/−55 V raw.

−40 V regulator (discrete)

R41 4k7 0.5W

D45 1N4755A 43V

R42 10k

C45 100u 63V

R43 1k

Q14 MPSA92

Q15 MJE350

R44 12

Q16 MPSA92

C46 100u 63V

C47 100n 100V

R45 10k 0.5W

J6 TO PREAMP +40/GND/GND/−40

Mirror: MPSA92 + MJE350. Output ~ −41.6 V.

Q12/Q15 (MJE340/MJE350) and Q2/Q5 (BD139/BD140) need small clip-on heatsinks.
BR1: GBJ2506 (25 A) — check the footprint against your part; give it a small heatsink.
T1: 400 VA toroid 2 x 20 VAC. T2: 15–30 VA 2 x 40 VAC. Centre taps go to GND (star point = C1–C4
Mains wiring (IEC inlet, fuse T3.15A/230V or T6.3A/115V, switch, 5 R NTC) is chassis-wired, not on t

Sheet: /
File: psu.kicad_sch

Title: Bass amp PSU board: +/−28 V, +/−40 V (discrete reg), +/−15 V (di

Size: A2	Date:	Rev: 1
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1